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# Lab # 15

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EL2003 – Computer Organization and Assembly Language



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Submitted To  
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## Task # 1

```
org 0x100

mov ax, 0
mov es, ax
mov ax, 0
mov es, ax
cli
mov word [es:9*4], kbisr
mov [es:9*4+2], cs
sti
mov ax, 0
mov es, ax
l1:
    mov ax, [count]
    push ax
    call printNum
    cmp byte[exit], 0
    je l1
mov ax, 0x4C00
int 0x21

kbisr:
    push ax
    push bx
    push es
    push di
    in al, 0x60
    cmp al, 0x1C
    je EnterP
        inc word[count]
        jmp leaveISR
EnterP:
    mov byte[exit], 1
leaveISR:
    mov al, 0x20
    out 0x20, al
    pop di
    pop es
    pop bx
    pop ax
    iret

printNum:
    push bp
    mov bp, sp
    push ax
    push bx
    push es
    push di
    mov ax, 0xB800
    mov es, ax
    mov ax, [bp+4]
    mov di, 2000
    mov bx, 10
printer:
```

```
    mov dx, 0
    div bx
    add dl, 0x30
    mov dh, 0x07
    mov word[es:di], dx
    mov word[es:3840], 'A'
    sub di, 2
    cmp ax, 0
    jne printer
pop di
pop es
pop bx
pop ax
pop bp
ret 2
```

```
exit: db 0
count: dw 0
```



## Task # 2

```
org 0x100
call clearScreen
mov ax, 0xB800
mov es, ax
mov ax, 0x0700
mov di, 160
mov si, msg
l1:
    lodsb
    stosw
    cmp al, 0
    jnz l1
push di
mov ax, string
push ax
call inputString
mov ah, 0x07
mov di, 320
mov si, msg2
l2:
    lodsb
    stosw
    cmp al, 0
    jnz l2
push di
mov si, string
push si
call printRevStr

mov ax, 0x4C00
int 0x21

inputString:
    push bp
    mov bp, sp
    push ax
    push bx
    push es
    push di
    push si
    mov ax, 0xB800
    mov es, ax
    mov si, [bp+4]
    mov di, [bp+6]
    mov bx, 0
    input:
        mov ax, 0
        int 0x16
        cmp ax, 0x1C0D
        jz endInput
        cmp ax, 0x0E08
        jnz si1
        cmp bx, 0
        jz si2
        dec bx
```

```

        sub di, 2
        mov byte[si+bx], 0x20
        mov word[es:di], 0x0720
        jmp si2
    si1:
        mov byte[si+bx], al
        mov ah, 0x07
        mov word[es:di], ax
        add di, 2
        inc word[size]
        inc bx
    si2:
        cmp bx, 30
        jnz input
endInput:
    mov byte[si+bx], 0
    pop si
    pop di
    pop es
    pop bx
    pop ax
    pop bp
    ret 2

```

printRevStr:

```

    push bp
    mov bp, sp
    push es
    push ax
    push cx
    push di
    push si

```

countingCharacters:

```

    push ds
    pop es
    mov si, [bp+4]
    mov di, [bp+4]
    mov ax, 0

    repne scasb
    mov cx, di
    sub cx, si

```

```

    mov ax, 0xB800
    mov es, ax
    mov ax, 0x072A
    mov di, cx
    dec di
    shl di, 1
    add di, [bp+6]
    mov si, [bp+4]
    sub di, 4

```

printer:

```

    cld
    lodsb
    std
    stosw

```

loop printer

```
mov al, 0
pop si
pop di
pop cx
pop ax
pop es
```

```
pop bp
ret 4
```

clearScreen:

```
push ax
push cx
push es
push di
mov ax, 0xB800
mov es, ax
mov di, 0
mov cx, 2000
mov ax, 0x0720
cld
rep stosw
pop di
pop es
pop cx
pop ax
ret
```

string: db ' ', 0

size: dw 0

msg: db 'Enter a String:', 0

msg2: db 'Reversed String: ', 0

```
DOSBox 0.74, Cpu speed: 3000 cycles, Frameskip 0, Program: DOSBOX
Enter a String: Hello World, What a nice day i
Reversed String: i yad ecin a tahW ,dlrow olleH

D:\>t i
```